	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 **Computer Science & Engineering**

INTERNALS-1
Scheme and Solutions

Course Code: 23CS5PEAAM	**Course Title: Advanced Algorithms**	
Semester: 5	**Maximum Marks: 40**	**Date:- 5-11-2024**
Faculty Handling the Course:	**Dr. GRP, Dr. BSR, NM, PP**	
Instructions: Internal choice provided in Part C.		

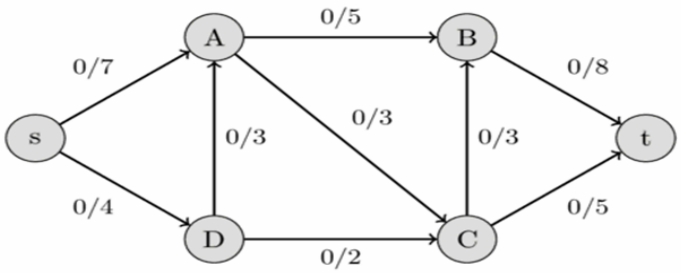
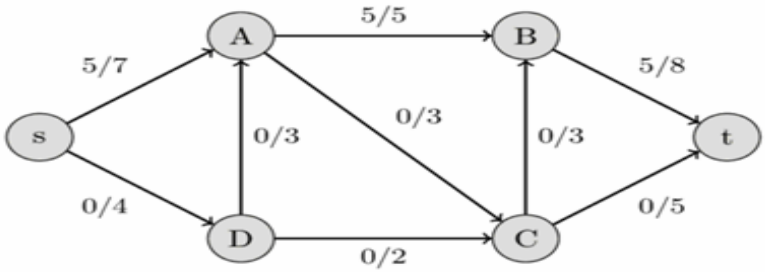
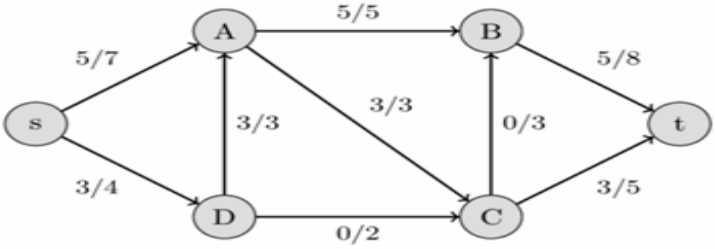
PART-A

Total 5 Marks (No Choice)

Q.No.	Question
1.	**Outline any 3 characteristics of Dynamic programming. How is it different from a Greedy approach?** **Any 3 characteristics——---------- 3 Marks** **Difference ——---------- 2 Marks** ➢ Dynamic programming is commonly used to solve optimization problems. ➢ A problem exhibits optimal substructure if an optimal solution to the problem can be constructed from optimal solutions of its subproblems. This means that you can solve larger problems by combining the solutions of smaller ones. ➢ DP is particularly useful when the problem can be broken down into subproblems that are reused multiple times. Instead of solving the same subproblem repeatedly, DP solves each subproblem once and stores its result, usually in a table or array, for future reference. The greedy approach makes the best choice at each step with the hope of finding a global optimum solution. Whereas Dynamic programming breaks down a problem into smaller subproblems and solves each subproblem only once, storing its solution. Greedy technique selects the locally optimal solution at each stage without considering the overall effect on the solution whereas Dynamic programming technique uses the results of solved subproblems to build up a solution to the larger problem.

PART-B

Total 15 Marks (No Choice)

Q.No.	Question
2 a)	Apply Ford Fulkerson algorithm to find the maximum flow that can flow from source S to sink t for the following flow network.  **S-A-B-T : 5**  **S-D-A-C-T : 3**  **S-D-C-B-T : 1** **S A D C T : 1** **Maximum flow : 10**
	Given the three operation: replace, insert and delete. Find the minimum number of operations

2 b)

required to convert following string 1 to string 2 using dynamic programming technique:

String 1: "elephant" **String 2: "relevant"**. Problem 4 Marks

		E	L	E	P	H	A	N	T
	0	1	2	3	4	5	6	7	8
R	1	1	2	3	4	5	6	7	8
E	2	1	2	2	3	4	5	6	7
L	3	2	1	2	3	4	5	6	7
E	4	3	2	1	2	3	4	5	6
V	5	4	3	2	2	3	4	5	6
A	6	5	4	3	3	3	3	4	5
N	7	6	5	4	4	4	4	3	4
T	8	7	6	5	5	5	5	4	3

Total minimum number of Operations required is 3 (1 Mark)

2 c)

Analyze and mention whether the following two DNA's belongs to same group or not. If sequence of size more than 4 is found, then they belong to the same group otherwise they not belong to the same group: **S1= "ABCBDAB "** **S2= "BDCABA"**

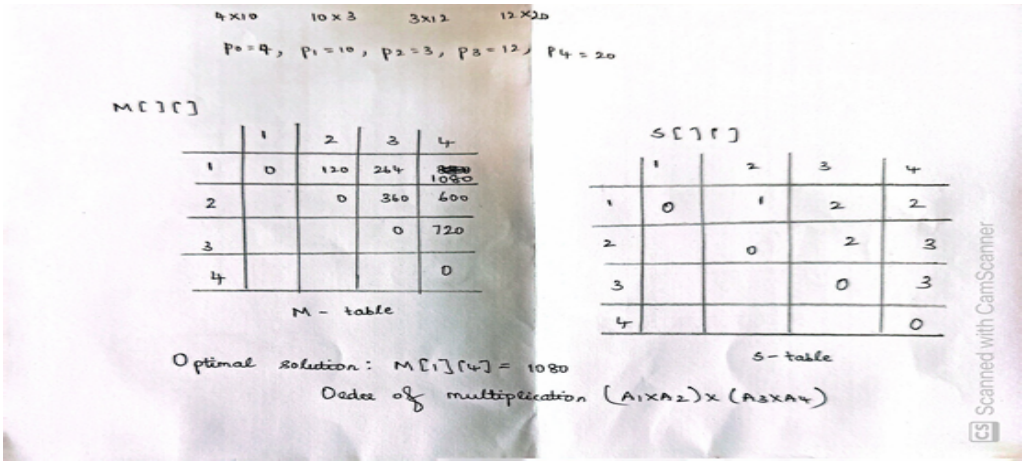
	'\0'	**A**	**B**	**C**	**B**	**D**	**A**	**B**
'\0'	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
B	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**
D	**0**	**0**	**1**	**1**	**1**	**2**	**2**	**2**
C	**0**	**0**	**1**	**2**	**2**	**2**	**2**	**2**
A	**0**	**1**	**1**	**2**	**2**	**2**	**3**	**3**
B	**0**	**1**	**2**	**2**	**3**	**3**	**3**	**4**
A	**0**	**1**	**2**	**2**	**3**	**3**	**4**	**4**

PART- C

Total 20 Marks

The common sub-sequence is of size 4 hence the DNA's belong to different groups.

Problem 3 Marks + Sub sequence size 1 mark + Conclusion 1 Mark

Q.No.	Question
3 a)	Justify how Matrix-Chain multiplication using Dynamic Programming technique reduces number of multiplications required compare to Naïve method for the following data: **A1 of order 4X10** **A2 of order 10X3** **A3 of order 3X12** **A4of order 12X20** Design an algorithm for the same and derive the time complexity of this algorithm. **Solution:** **Problem solving with all steps: 6 marks** **Final order: 1 mark**  **Algorithms: 2 marks** **Time complexity: 1 mark** MATRIX-CHAIN-ORDER(p) 1 $n \leftarrow length[p] - 1$ 2 **for** $i \leftarrow 1$ **to** n 3 **do** $m[i, i] \leftarrow 0$ 4 **for** $l \leftarrow 2$ **to** n ▷ l is the chain length. 5 **do for** $i \leftarrow 1$ to $n - l + 1$ 6 **do** $j \leftarrow i + l - 1$ 7 $m[i, j] \leftarrow \infty$ 8 **for** $k \leftarrow i$ **to** $j - 1$ 9 **do** $q \leftarrow m[i, k] + m[k + 1, j] + p_{i-1} p_k p_j$ 10 **if** $q < m[i, j]$ 11 **then** $m[i, j] \leftarrow q$ 12 $s[i, j] \leftarrow k$ 13 **return** m and s PRINT-OPTIMAL-PARENS(s, i, j) 1 **if** $i = j$ 2 **then** print "A"$_i$ 3 **else** print "(" 4 PRINT-OPTIMAL-PARENS($s, i, s[i, j]$) 5 PRINT-OPTIMAL-PARENS($s, s[i, j] + 1, j$) 6 print ")"

OR

3 b) Suppose there is a building with 7 floors and you are given 3 glass pieces. Determine the minimum number of experiments required to find the threshold floor from which you can drop a glass piece without breaking. Write an algorithm for the same and find its time complexity.

Solution:

Problem solving with all steps: 7 marks

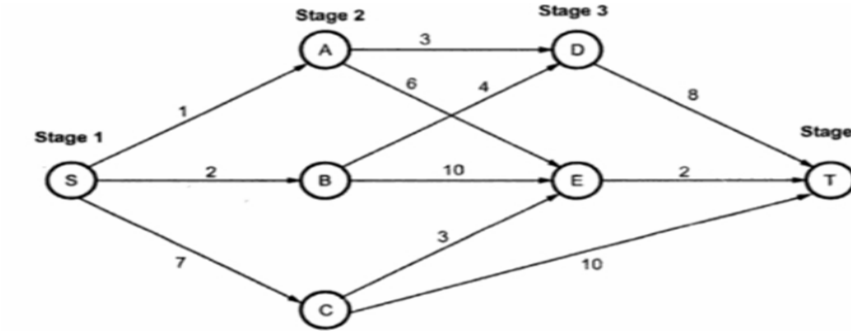
eggs / floor	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	1	2	2	3	3	3	4
3	0	1	2	2	3	3	3	3

Algorithms: 2 marks

```
int max(int a, int b)
{    return (a > b) ? a : b; }
 int eggDrop(int n, int k)
{
   if (k == 1 || k == 0)
     return k;
    if (n == 1)
     return k;
   int min = INT_MAX, x, res;
     for (x = 1; x <= k; x++)
  {
     res = max( eggDrop(n - 1, x - 1), eggDrop(n, k - x));
     if (res < min)    min = res;
  }
  return min + 1;
}
```

Time complexity: 1 mark

4a) Find the minimum cost involved in reaching from node S to node T for the following multistage graph using forward approach. Design an algorithm for the same.



Solution

Steps ---------- 5 Marks

Final Answer ---------- 1 Mark

Algorithm ---------- 4 Marks

Multistage Graph - Forward Approach

Vertex	S	A	B	C	D	E	T
Cost	9	8	12	5	8	2	0
d	A	E	D/E	E	T	T	T

I

$Cost(T) = 0$

$d(T) = 0$

II

$Cost(D) = Cost(T) + c(D,T) = 0+8 = 8$

$Cost(E) = Cost(T) + c(E,T) = 0+2 = 2$ ←

III

$Cost(A) = \min \{ Cost(D) + c(A,D) = 8+3 = 11,\ Cost(E) + c(A,E) = 2+6 = 8 \}$ ←

$Cost(B) = \min \{ Cost(D) + c(B,D) = 8+4 = 12 ←,\ Cost(E) + c(B,E) = 2+10 = 12 \}$ ←

$Cost(C) = \min \{ Cost(E) + c(C,E) = 2+3 = 5 ←,\ Cost(T) + c(C,T) = 0+10 = 10 \}$

IV

$Cost(S) = \min \{ Cost(A) + c(S,A) = 8+1 = 9 ←,\ Cost(B) + c(S,B) = 12+2 = 14,\ Cost(C) + c(S,C) = 5+7 = 12 \}$

∴ min Cost = 9

path = S → A → E → T

```
1   Algorithm FGraph(G, k, n, p)
2   // The input is a k-stage graph G = (V, E) with n vertices
3   // indexed in order of stages. E is a set of edges and c[i, j]
4   // is the cost of ⟨i, j⟩. p[1 : k] is a minimum-cost path.
5   {
6        cost[n] := 0.0;
7        for j := n − 1 to 1 step −1 do
8        { // Compute cost[j].
9             Let r be a vertex such that ⟨j, r⟩ is an edge
10            of G and c[j, r] + cost[r] is minimum;
11            cost[j] := c[j, r] + cost[r];
12            d[j] := r;
13       }
14       // Find a minimum-cost path.
15       p[1] := 1; p[k] := n;
16       for j := 2 to k − 1 do p[j] := d[p[j − 1]];
17  }
```

OR

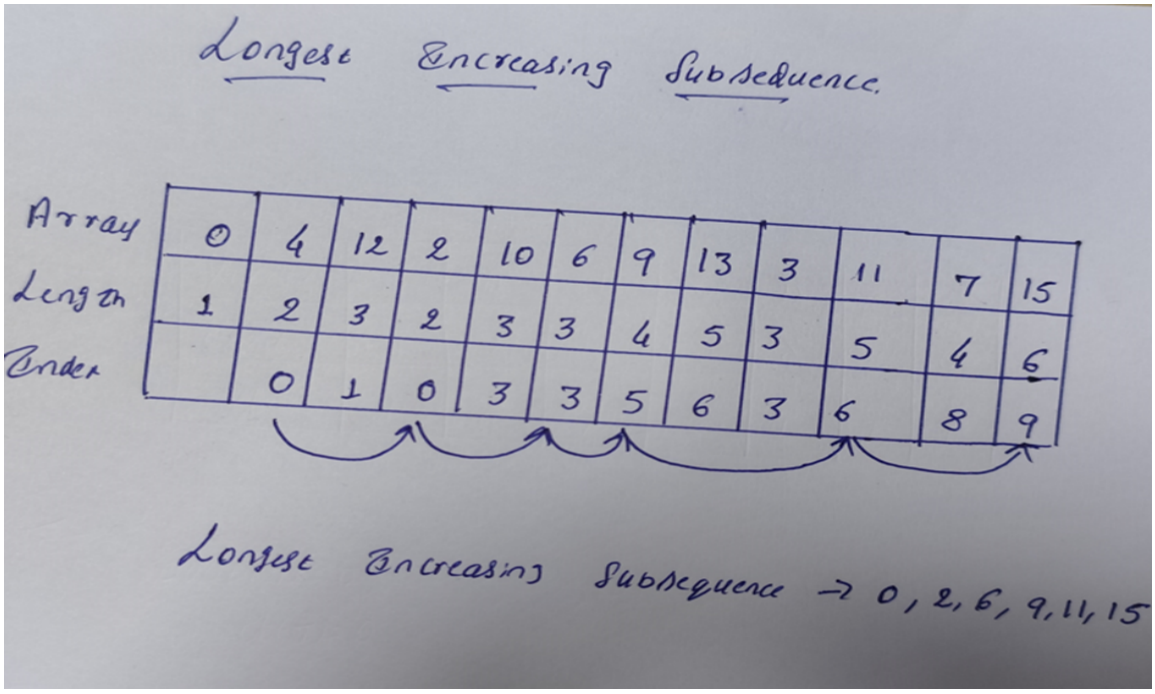
4b) Find longest increasing subsequence for the following sequence of integers: 0,4,12,2,10,6,9,13,3,11,7,15. Design an algorithm and find the time complexity of this algorithm.

Solution

Steps ---------- 5 Marks

Final Answer ---------- 1 Mark

Algorithm ---------- 4 Marks



for j=1 to n
do L[j]←1
P[j]←0
for i=1 to j-1 do
if (si L[j]) then
P[j] ←i L[j] ←L[i]+1
endfor
endfor
• Now find j such that L[j] is largest and walk backwards through P[j] pointers to find the sequence.

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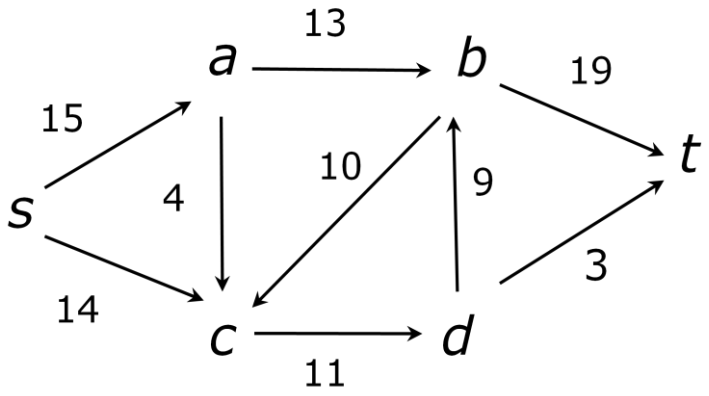
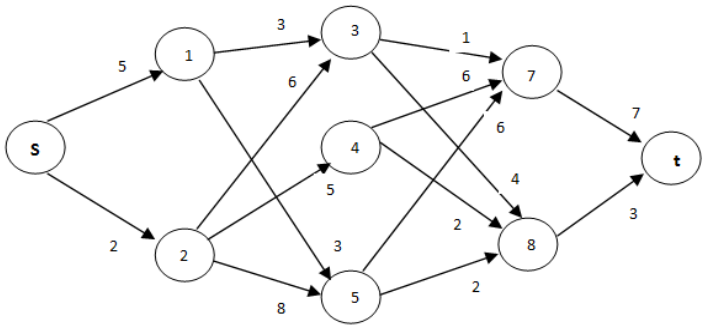
PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO No.	Level
1a	What are steps used in solving problems in the dynamic programming approach?	**5M**	**1**	**4**

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO No.	Level
2a	Apply Ford Fulkerson Method to find maximum flow in the below flow network. 	**5M**	**2**	**3**
2b	Find shortest path from source to destination in the multistage graph given below using Dynamic programming with forward approach. 	**5M**	**2**	**3**
2c	Apply Dynamic programming to find threshold floor in Egg dropping puzzle, assuming n=3 eggs and k=7 floors.	**5M**	**2**	**3**

PART- C

Total 20 Marks (Choice)

<table>
<tr><th>No.</th><th>Question</th><th>Marks</th><th>CO No.</th><th>Level</th></tr>
<tr><td>3a</td><td>Design an algorithm to find solution to rod cutting problem using dynamic programming. Also apply the same to solve below instance.

Rod length=7

<table>
<tr><td>Length i</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr>
<tr><td>Price Pi</td><td>2</td><td>5</td><td>8</td><td>10</td><td>13</td><td>16</td><td>17</td><td>22</td><td>24</td><td>30</td></tr>
</table>
</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>3b</td><td>Design an algorithm to find order in which matrices are to be multiplied so as to minimize the number of multiplications using dynamic programming. Also apply the same to solve below instance.

A1 *A2*A3*A4*A5

A1=8*4 A2=4*6 A3=6*5 A4=5*4 A5=4*7</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td>4a</td><td>Design pseudocode/program to find longest common sequence between two strings using dynamic programming. Also apply the same to solve below instance.

S1=bacbff and S2=dabeab</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4b</td><td>Design pseudocode/program to find longest increasing subsequence using dynamic programming. Trace the algorithm for the following input sequence and find longest increasing subsequence in T={4, 2, 5, 3, 9, 7, 8, 10, 6}.</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
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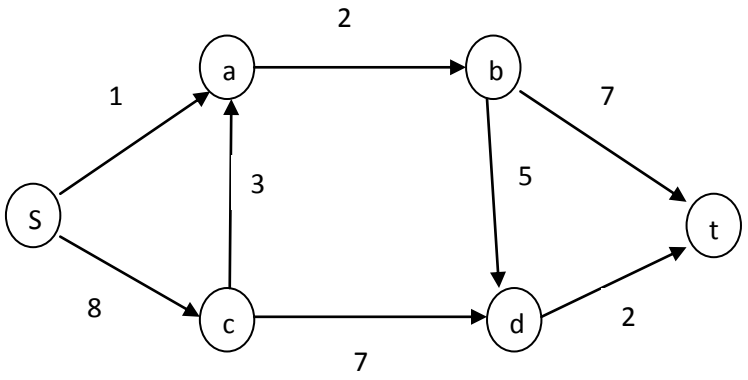
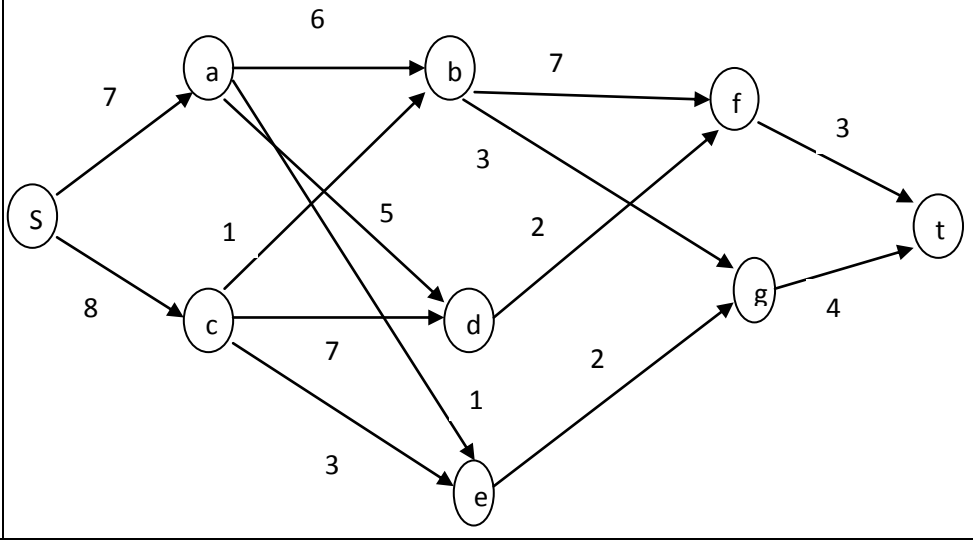
PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO No.	Level
1a	Differentiate between bottom up and top down design approaches with an example.	**5M**	**1**	**4**

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO No.	Level
2a	Apply Ford Fulkerson Method to find maximum flow in the below flow network. 	**5M**	**2**	**3**
2b	Find shortest path from source to destination in the multistage graph given below using Dynamic programming with forward approach. 	**5M**	**2**	**3**

2c	Apply Dynamic programming to find threshold floor in Egg dropping puzzle, assuming n=4 eggs and k=4 floors.	**5M**	**2**	**3**

PART- C

Total 20 Marks (Choice)

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Rod length=6

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A1 *A2*A3*A4*A5

A1=8*3 A2=3*6 A3=6*4 A4=4*5 A5=5*4</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td>4a</td><td>Design pseudocode/program to find longest common sequence between two strings using dynamic programming. Also apply the same to solve below instance.

S1=ABCDAB and S2=BDCABA</td><td>10M</td><td>2,3</td><td>2,3</td></tr>
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